NLPLab at ArGuard Shared Tasks: Balancing Confidence Scores for Fine-Grained Classification of Unsafe Arabic Prompts Across Harm Domains

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Abstract

This paper presents our system for ArGuard 2026 Subtask B2, which classifies unsafe Arabic prompts into one of seven harm domains. Our system combines four complementary components: a fine-tuned CAMeLBERT-Mix model, a modified-Huber stochastic gradient descent classifier, a character-level nearest-neighbor model, and a character class-centroid similarity model. On the development set, we achieved a Macro-F1 score of 0.916 and an accuracy of 0.923. On the test set, the proposed model achieved an accuracy of 0.766, a macro precision of 0.760, a macro recall of 0.763, and a Macro-F1 of 0.751. The results show that combining contextual transformer representations, lexical features, and retrieval-based similarity signals can improve fine-grained harmful-prompt classification.

1 Introduction

Text classification assigns predefined labels to textual inputs and has evolved from rule-based and feature-engineered methods to pretrained transformer models that capture contextual language representations (Abiola et al., 2025, 2026; Adebanji et al., 2022, 2024; Ojo et al., 2020, 2023, 2024). Fine-grained harm classification requires assigning each unsafe prompt to a single category among several closely related harm types. This is inherently difficult because harmful categories often overlap. For example, a prompt may contain bullying, harassment, deception, and threats of physical harm at the same time. Similarly, a self-harm-related prompt may also involve emotional manipulation, abuse, or threats toward another person. Despite these overlaps, the task requires the system to assign only one label to each unsafe prompt. This creates difficult cases, especially when prompts are short, indirect, vague, or written in informal language.

Arabic introduces additional challenges. Prompts may be written in Modern Standard Arabic, regional dialects, Arabic-English mixed text, informal spelling, or non-standard punctuation. Arabic is also morphologically rich, and harmful language may differ across countries, dialects, and online communities. These factors make it difficult for a single model or text representation to perform consistently across all harm categories. Previous work on Arabic safety and offensive-language detection has also highlighted the effects of dialect variation and the difficulty of fine-grained harmful-content classification (Mubarak et al., 2020; Zaghouani et al., 2026; Mubarak et al., 2025).

ArGuard 2026 focuses on harmful-content detection and the fine-grained classification of unsafe Arabic prompts (Alam et al., 2026; Kmainasi et al., 2026). In Track B, the task focuses on detecting harmful Arabic prompts for LLMs; Subtask B2 requires assigning each unsafe prompt to one of seven fine-grained harm domains: *Bully*, *FraudandDeception*, *adult_content*, *harassment*, *harm-to-others*, *hate_speech*, or *self-harm*. Our approach is based on the idea that no single model type is equally effective for all seven categories. Transformer models can capture semantic meaning and contextual relationships. Sparse linear models can identify highly discriminative word and character patterns. Retrieval-based methods can compare a new prompt with similar labeled examples from the training data. We therefore developed a heterogeneous ensemble that combines semantic, lexical, and similarity-based signals.

We combined four components, including a fine-tuned CAMeLBERT-Mix classifier, an averaged stochastic gradient descent classifier trained with modified-Huber loss, a character-level nearest-neighbor classifier, and a character class-centroid similarity model. We calibrated the model probabilities using temperature scaling, assigned equal weights to the selected models, and applied small

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 437–442
October 28-29, 2026 ©2026 Association for Computational Linguistics

Table 1: Harm-domain labels in ArGuard 2026 Subtask B2.

No.	Label
1	*Bully*
2	*FraudandDeception*
3	*adult_content*
4	*harassment*
5	*harm-to-others*
6	*hate_speech*
7	*self-harm*

Table 2: Data splits used in the experiments.

Split	Safe	Unsafe	Total
Training	3,391	13,384	16,775
Development	475	1,928	2,403
Test	-	5,979	5,979

class-specific probability adjustments. In a few high-confidence cases, the system used retrieval-based predictions when closely matching training examples provided strong evidence. Our main contributions include a multi-view preprocessing pipeline for Arabic harmful prompts, a comparison of transformer-based and sparse linear classification methods, two character-based retrieval components for identifying similar labeled examples, and a calibrated heterogeneous ensemble for seven-way harmful-prompt classification.

Figure 1 presents an overview of our system.

2 Data and Preprocessing

2.1 Task and Data

Subtask B2 requires assigning each unsafe Arabic prompt to one of seven mutually exclusive harm domains shown in Table 1. The released data contained 16,775 training examples and 2,403 development examples. The training set included 13,384 unsafe and 3,391 safe prompts, while the development set included 1,928 unsafe and 475 safe prompts. Since B2 evaluates fine-grained classification only for unsafe inputs, model selection was performed on the 1,928 unsafe development prompts. The test set contained 5,979 unsafe prompts.

Although safe prompts were not evaluated directly, we retained them as low-weight auxiliary examples because they were linked to harm-related topic labels. The best LinearSVC setting assigned them a weight of 0.12. The classes were imbalanced, with *adult_content* as the largest test class with 1,311 examples and *hate_speech* as the smallest with 351 examples.

2.2 Preprocessing and Features

We created original, normalized, and simplified versions of each prompt. The original text preserved lexical and stylistic information, while the normalized text reduced variation caused by informal writing and Arabic orthographic differences. We replaced URLs, email addresses, usernames, and hashtags with special markers; removed diacritics and tatweel; normalized common Arabic letter variants; and reduced character repetitions of three or more occurrences to two. The simplified representation replaced digit sequences with `NUM` and removed non-word symbols. When available, dialect information was appended to the input. Sparse models used three TF-IDF feature groups: normalized word 1-3-grams, character-boundary 2-7-grams, and raw character 3-6-grams. The maximum numbers of features were 140,000, 280,000, and 140,000, respectively. All representations used `min_df=2`, sublinear term-frequency scaling, L2 normalization, and 32-bit floating-point values. The final sparse matrix contained 299,867 features. Word features captured lexical patterns, while character features improved robustness to spelling variation, dialectal forms, punctuation, and partially obfuscated expressions.

3 System Design

Our system combines transformer models, sparse lexical classifiers, and character-based retrieval methods. Transformers capture contextual meaning, sparse models identify discriminative word and character patterns, and retrieval components compare a prompt with similar labeled training examples. We evaluated each branch on the unsafe development set, calibrated its probabilities, and selected complementary models for the final ensemble.

3.1 Sparse and Retrieval Models

We evaluated three sparse classifiers using the combined TF-IDF features. LinearSVC achieved the strongest sparse result, with $C = 0.65$ and safe-example weight $w_{\text{safe}} = 0.12$, obtaining 0.880318 development Macro-F1. An averaged `SGDClassifier` with modified-Huber loss and L2 regularization achieved 0.856061 Macro-F1 using $\alpha = 10^{-5}$. Complement Naive Bayes achieved 0.849654 Macro-F1 with $\alpha = 0.15$.

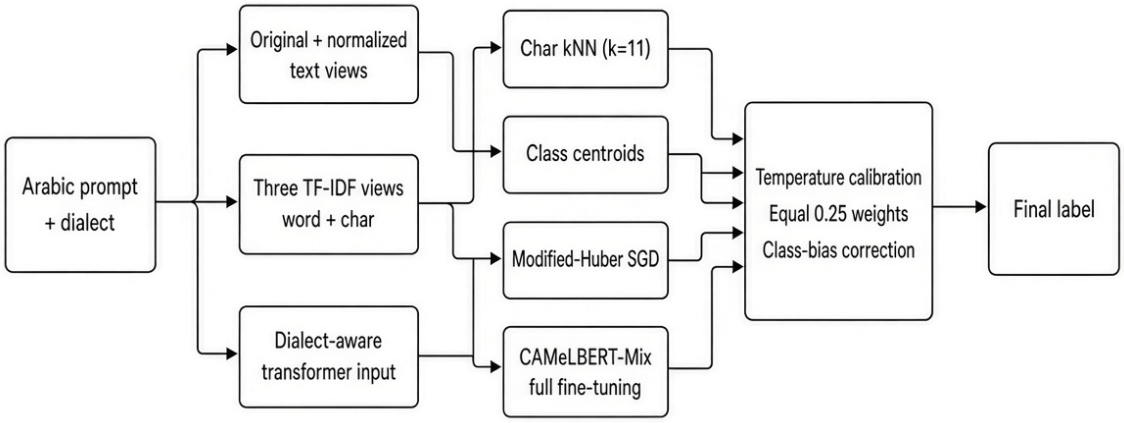


Figure 1: System architecture of the proposed heterogeneous ensemble

We also used two character-based retrieval components. Character kNN retrieved similar training prompts using cosine similarity over character TF-IDF vectors and predicted classes through similarity-weighted voting. The best configuration used $k = 11$ neighbors and weighting exponent $p = 2$, achieving 0.767532 Macro-F1. The class-centroid model represented each class with a normalized character-TF-IDF centroid and scored prompts by cosine similarity to these centroids. It achieved 0.655894 Macro-F1 but produced complementary predictions that improved the ensemble.

3.2 Transformer Models

We fine-tuned MARBERT and CAMeLBERT-Mix, which support Modern Standard Arabic and dialectal Arabic (Abdul-Mageed et al., 2021; Inoue et al., 2021). Models used a maximum input length of 192 tokens and were trained for up to five epochs with early stopping after two epochs without development improvement. The training batch size was 8, the evaluation batch size was 24, and gradients were accumulated over two steps. We used encoder and classification-head learning rates of 1.6×10^{-5} and 6.0×10^{-5}, respectively, with weight decay of 0.01, label smoothing of 0.06, an 8% warmup ratio, gradient clipping at 1.0, and layer-wise learning-rate decay of 0.92. Normalized-text augmentation was applied with probability 0.25 during training. At inference time, predictions from original and normalized text were combined with weights of 0.78 and 0.22. MARBERT achieved 0.883974 development Macro-F1 at epoch 3, while CAMeLBERT-Mix achieved the best individual score of 0.889302 at epoch 4. Both models were subsequently refit for one additional epoch on the combined training and development data using learning rates reduced to 35% of their original values.

3.3 Ensemble and Adjustments

Each branch was temperature-calibrated on the unsafe development set using the temperature that minimized log loss. We then used greedy selection, adding a model only when it improved development Macro-F1 by at least 0.0005. The final ensemble included CAMeLBERT-Mix, modified-Huber SGD, character kNN, and class-centroid similarity.

$$p_{\text{ens}} = 0.25p_{\text{CAMeL}} + 0.25p_{\text{SGD}} + 0.25p_{\text{kNN}} + 0.25p_{\text{centroid}}. \quad (1)$$

The equal-weight ensemble achieved 0.910065 development Macro-F1. MARBERT, LinearSVC, and Complement Naive Bayes were excluded because they did not further improve the ensemble. We applied small class-specific biases to the ensemble log probabilities. This increased Macro-F1 from 0.910065 to 0.916048 and accuracy from 0.9180 to 0.9238. The final inference procedure also assigned labels to 33 exact normalized matches with unambiguous training labels and applied only two very-high-similarity retrieval overrides.

To make the probability adjustment reproducible, we first computed $z_{ic} = \log(\max(p_{ic}, 10^{-9}))$ for example i and class c, and then predicted with $\arg\max_c(z_{ic} + b_c)$. Starting from a zero vector, we performed two coordinate-descent passes over the classes in label order and tested each coordinate against the fixed grid $\{-0.30, -0.18, -0.08, 0, 0.08, 0.18, 0.30\}$. A candidate was retained only when it improved de-

Table 3: Development Macro-F1 on the unsafe development set

System	Macro-F1
CAMeLBERT-Mix	0.889302
MARBERT	0.883974
LinearSVC	0.880318
Modified-Huber SGD	0.856061
ComplementNB	0.849654
Character kNN	0.767532
Class centroid	0.655894
Equal-weight four-branch ensemble	0.910065
Ensemble + class adjustment	**0.916048**

velopment Macro-F1 after subtracting the regularizer $0.0015\,\mathrm{mean}(|\mathbf{b}|)$; the entire adjustment was accepted only if its unregularized Macro-F1 exceeded the unadjusted ensemble by at least 0.0025. The selected biases, in the order *Bully*, *FraudandDeception*, *adult_content*, *harassment*, *harm-to-others*, *hate_speech*, and *self-harm*, were $[0.00, 0.18, -0.30, 0.18, -0.08, 0.00, 0.08]$. A softmax was applied after adding these biases when probabilities and confidence values were required; the argmax labels are unchanged by that final normalization.

4 Results

4.1 Development Results

Table 3 presents the development results. CAMeLBERT-Mix was the strongest individual model, followed by MARBERT and LinearSVC. Although the retrieval models had lower standalone scores, they improved the ensemble because they captured complementary patterns. The final development system achieved 0.9238 accuracy. Class-level F1 scores ranged from 0.8537 for *harm-to-others* to 0.9990 for *adult_content*. The *Bully* class achieved 0.8894 F1 on development but showed the largest decline on the test set.

Table 4 reports the component-wise incremental ablation recorded during greedy selection. Each row adds one branch to the preceding configuration, so the score change isolates that component's marginal contribution conditional on the branches already selected. The two retrieval systems are weak in isolation, but character kNN adds 0.003039 Macro-F1 after CAMeLBERT-Mix and SGD, and the centroid branch adds a further 0.001153. These gains support complementarity without implying that the retrieval branches are strong standalone classifiers.

Table 4: Incremental ablation of the four final ensemble components on the unsafe development set. Δ is relative to the preceding row.

Components included	Macro-F1	Δ
CAMeLBERT-Mix	0.889302	–
+ Modified-Huber SGD	0.905873	+0.016571
+ Character kNN	0.908912	+0.003039
+ Class centroid	0.910065	+0.001153

Table 5: Result on test set. Precision and recall are macro-averaged.

System	Acc.	Prec.	Rec.	Macro-F1
Baseline	0.702	0.685	0.707	0.687
Our System	**0.766**	**0.760**	**0.763**	**0.751**

4.2 Test Results

Our system achieved 0.766 accuracy, 0.760 macro precision, 0.763 macro recall, and 0.751 Macro-F1 on the test set. This corresponds to 4,580 correct predictions out of 5,979 prompts. The system improved on the reported baseline by 0.064 Macro-F1.

Table 6 reports class-level test results. The model performed best on *adult_content*, with 0.963 F1, and performed strongly on *hate_speech*, with 0.806 F1. The weakest category was *Bully*, which obtained 0.768 precision but only 0.415 recall, resulting in 0.539 F1. This indicates that many bullying prompts were assigned to related categories such as harassment or harm to others.

Macro-F1 decreased by 0.165048 from development to test, mainly because of the decline in *Bully* performance from 0.8894 to 0.539 F1. The system predicted 491 bullying prompts, while the gold-label analysis reported 908 examples. This result highlights the difficulty of distinguishing bullying from closely related categories and suggests that the category boundary differed between the development and test sets. In addition, extensive reuse of the same development labels for hyperparameter selection, probability calibration, greedy ensemble construction, class-bias fitting, and override validation may have overfit this split and inflated the development estimate, thereby contributing to the observed test degradation.

5 Conclusion

We present our system for ArGuard 2026 Subtask B2, a seven-class classification task for unsafe Arabic prompts. The system evaluated sparse lin-

Table 6: Class-level test performance.

Class	Support	Prec.	Rec.	F1
Bully	908	0.768	0.415	0.539
Fraud/deception	689	0.632	0.787	0.701
Adult content	1,311	0.965	0.960	0.963
Harassment	674	0.713	0.864	0.781
Harm to others	1,166	0.701	0.719	0.710
Hate speech	351	0.818	0.795	0.806
Self-harm	880	0.722	0.799	0.758

ear classifiers, character-based retrieval methods, MARBERT, and CAMeLBERT-Mix. The final system calibrated model probabilities, combined the selected components using equal weights, and applied conservative class-specific adjustments. The final approach improved development Macro-F1 from 0.910065 to 0.916048. On the test set, the system achieved 0.766 accuracy, 0.760 macro precision, 0.763 macro recall, and 0.751 Macro-F1. The results show that heterogeneous ensembles are useful for Arabic harmful-prompt classification because transformer, sparse, and retrieval-based models capture different types of information. However, the substantial performance drop on the test set, especially for the *Bully* category, shows that fine-grained harm classification remains sensitive to changes in category boundaries, Arabic varieties, and data distributions.

6 Ethical Considerations

Fine-grained harmful-prompt classification involves sensitive content, including self-harm, threats, harassment, hate speech, adult content, fraud, and violence. Researchers should protect this data with access controls and use it only for research and safety purposes to reduce harm, misuse, and unnecessary exposure.

7 Limitations

The main model selection and post-processing decisions were based on one development split. The component study followed the recorded greedy search and did not test every possible ensemble combination. Exact match and similarity rules affected only a few examples, but their thresholds were also selected using the same development data. The test results show that performance may change under shifts in data or annotation boundaries, and the findings may not apply directly to other Arabic varieties, domains, or harm categories.

Acknowledgments

This work was supported in part by the Mexican Government through Secretaría de Ciencia, Humanidades, Tecnología e Innovación (SECIHTI), Comisión de Operación y Fomento de Actividades Académicas del IPN (COFAA-IPN), and Secretaría de Investigación y Posgrado of the Instituto Politécnico Nacional, Mexico, under grants 20260569, 20260626, and 20260643.

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